Marc-Aurèle **RIVIÈRE**

Data Scientist

Trondheim, Norway

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Building software and analyzing data for 7 years, with a focus on statistical modeling, R & Shiny.

Experience

R/Shiny Developer — Remote

Data Champ' — 2024 - 2025

Delivered data products end-to-end for clients across multiple industries, from requirements and architecture through deployment and iteration, interfacing with both technical and non-technical stakeholders. Projects included:

- An investment analytics platform for France's development agency (AFD), to monitor their portfolio of billions in international development funding across thousands of projects, with multi-database integration, interactive geospatial visualizations, and automated impact reporting.
- An enterprise-scale energy Measurement & Verification API for the Belgian Government, automating standard and IPMVP methods for energy consumption modeling, statistical analyses, multi-platform data integration, and estimating the impact of renovations on energy consumption.
- A comprehensive HR management system for the AFD to track thousands of employees across multiple branches, featuring recruitment campaign management, and role-based workflows.
- A modular package for a branch of the French government, to integrate multiple sources of heterogeneous geo-spatial data, process them into spatio-temporal indicators at various levels of spatial granularity, and archive them in a specially-designed DB.

Data Science Consultant — Remote

Freelance — 2023 - 2024

Sales/Marketing:

- Developed an agentic prospect discovery tool with social media scraping, multi-CRM integration, and lead-database enrichment
- Built a personalized outreach pipeline with RAG over company offerings, and A/B analysis against traditional template-based emails

Bioinformatics:

- Data analysis for 4 published neuroscience papers: hierarchical GLMMs across heterogeneous data types (IHC, imaging, qPCR, behavioral); AR1 structures for time-varying repeated measures; multiple candidate models with PPC & AIC-based selection
- Created app to allow researchers to visualize and analyze their data, with automatic enrichment from NCBI (R/Shiny)
- Built interactive dashboards to showcase results for select publications (R, Quarto, JS)

Research Engineer — Rouen, France

LITIS — 2021 - 2023

- Designed & developed an Augmented Reality platform with motion tracking to evaluate and train non-visual navigation (Unity/C#)
- Designed a wearable vibrotactile belt and an interactive haptic tablet for haptic-based navigation for blind people (Java/Arduino)
- Designed psychophysics experiments to evaluate the impact of spatial cues on navigation and analyzed the resulting data (R)
- Developed and tested camera-based solutions for indoor localization (Python/C++)
- Formulated & wrote a 600k€ research grant proposal (ANR project "SAM-Guide", 2021)
- Co-organized international and regional scientific conferences

Graduate Teaching Fellow — Rouen, France

University of Rouen — 2017 - 2020

Gave university lectures & practicals on Web Development (JS), GUI design (Java), Image Processing/Computational Imaging (Python)

Research Engineer Intern — San Francisco, USA

SKERI — 2018

Worked on an indoor localization tool using Visual Inertial Odometry, particle filtering, and object detection (Python, Swift, OpenCV)

Research Engineer Intern — Grenoble, France

LPNC & GIPSA-Lab — 2016

Worked on an image-to-sound conversion app for blind image exploration (C++), iteratively improved based on users' performance

Education

PhD - Biomedical Engineering
Rouen, France

Normandy University (unfinished - COVID)
2017 - 2020

- Design, develop and evaluate wearable haptic devices to assist blind people with navigation tasks in augmented reality (Java)
- Design, execute, and analyze multiple psychophysics experiments with human subjects (R)
- Implement algorithms for real-time localization and mapping using embedded/mobile sensors (Python/C++)

MSc. - Computational Neurosciences
Grenoble, France

PHELMA - INPG
2015 - 2016

- Fundamentals of Memory, Perception, Linguistics, Imaging (e.g. fMRI, EEG, MEG), and Robotics
- Experimental design, Statistics, Bayesian modeling, Deep Learning, Signal processing (MATLAB & R)

Skills

Programming	**R**, Python, SQL, JS, Java, C#, Stan
Frameworks & Tools	**Shiny**, **Plumber**, **Quarto**, Git, CI/CD
DBMS	**Postgres**, **DuckDB**, SQLite
DevOps / Infra	Ansible, Terraform, Docker
Languages	**French (Native)**, **English (Bilingual)**, Norwegian (A2/B1), Spanish (A2)

Publications

- Rodriguez-Duboc, A., Basille-Dugay, M., Debonne, A., **Rivière, M.-A.**, Vaudry, D., & Burel, D. (2023). Apnea of prematurity induces short and long-term development-related transcriptional changes in the murine cerebellum. *Current Research in Neurobiology*, 5, 100113.
- Faugloire, E., Lejeune, L., **Rivière, M.-A.**, & Mantel, B. (2022). Spatiotemporal influences on the recognition of two-dimensional vibrotactile patterns on the abdomen. *Journal of Experimental Psychology: Applied*, 28(3), 606–628.
- Coughlan, J.M., Biggs, B., **Rivière, M-A.**, Shen, H. (2020). An Audio-Based 3D Spatial Guidance AR System for Blind Users. In *Lecture Notes in Computer Science* (Vol. 12376, pp. 475–484). Springer.
- **Rivière, M-A.**, Gay, S., Romeo, K., Pissaloux, E., Bujacz, M., Skulimowski, P., & Strumillo, P. (2019). NAV-VIR: An audio-tactile virtual environment to assist visually impaired people. *Proceedings of the 9th International IEEE/EMBS Conference on Neural Engineering*, 1038–1041.
- **Rivière, M-A.**, Gay, S., Pissaloux, E. (2018). TactiBelt: Integrating Spatial Cognition and Mobility Theories into the Design of a Novel Orientation and Mobility Assistive Device for the Blind. In *Lecture Notes in Computer Science* (Vol. 10897, pp. 110–113). Springer.
- Gay, S., **Rivière, M-A.**, Pissaloux, E. (2018). Towards Haptic Surface Devices with Force Feedback for Visually Impaired People. In *Lecture Notes in Computer Science* (Vol. 10897, pp. 258–266). Springer.

Data analysis contributions

- Rodriguez-Duboc, A., Racine, C., Basille-Dugay, M., Vaudry, D., Gonzalez, B., & Burel, D. (2026). Innovative 3D-image analysis of cerebellar vascularization highlights angiogenic gene dysregulations in a murine model of apnea of prematurity. *The Cerebellum*.
- Rodriguez-Duboc, A., Velicer, S. T., & Kobro-Flatmoen, A. (2025). Bnip3 expression in the brain of an Alzheimer's disease rat model. *Brain Mechanisms*, 148–150, 202518.
- Leroux, S., Rodriguez-Duboc, A., Arabo, A., Basille-Dugay, M., Vaudry, D., & Burel, D. (2022). Intermittent hypoxia in a mouse model of apnea of prematurity leads to a retardation of cerebellar development and long-term functional deficits. *Cell & Bioscience*, 12(1), 148.

Packages

Argent
ma-riviere/Argent

R Package

A unified interface to build AI Agents using Large Language Models (LLMs) from multiple providers, supporting tool calling (functions, server tools, and MCP servers), multimodal inputs, and structured outputs.